QUALITY INFORMATION DOCUMENT

For
GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies
GLOBAL_OMI_TEMPSAL_sst_trend

Issue: 3.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	16/08/2019	All	New document	Simon Good	
1.1	10/09/2020		Addition of 2019	Simon Good	
1.2	05/08/2021		Addition of 2020	Simon Good	
2.0	16/06/2022	All	New template and addition of 2021	Simon Good	
3.0	16/05/2025	All	New document template, updated input dataset, climatology period and temporal extent	Simon Good, Matilde Brandt Kreiner	Copernicus Marine Product Management

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1) INTRODUCTION

a) Product Name

GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies and GLOBAL_OMI_TEMPSAL_sst_trend

b) Indicator Evaluation Method

These OMIs are based on daily, global climate sea surface temperature (SST) analyses generated by the Copernicus Climate Change Service (C3S). The data are available from the originators and from the Copernicus Marine Service (product SST-GLO-SST-L4-REP-OBSERVATIONS-010-024).

Analysis of the data was based on the approach described in Mulet et al. (2018). The processing steps applied were:

1. The daily analyses were averaged to create monthly means.
2. A climatology was calculated by averaging the monthly means over the period 1991 - 2020.
3. Monthly anomalies were calculated by differencing the monthly means and the climatology.
4. An area averaged time series (provided within the indicator GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies) was calculated by averaging the monthly fields over the globe, with each grid cell weighted according to its area.
5. The time series in each grid cell and the area averaged time series were each passed through the X11 seasonal adjustment procedure, which decomposes the time series into a residual seasonal component, a trend component and errors (e.g. Pezzulli et al., 2005). The trend component is a filtered version of the monthly time series. The trend component from the area averaged time series is included in indicator GLOBAL_OMI_TEMPSAL_sst_area_averaged_anomalies.
6. The slope of the trend component for the time series in each grid cell and the area averaged time series was calculated using a robust method (Sen 1968). The method also calculates the 95% confidence range in the slope.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
SST_GLO_SST_L4_REP_OBSERVATIONS_010_024	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-024.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SST-PUM-010-024.pdf

2) SCIENTIFIC VALIDATION

The basis of these indicators is the accuracy and stability over time of the source data from the C3S project. The quality of these products is discussed in the QuID for this product: https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-024.pdf. The median and robust standard deviation of difference between the sea surface temperature analyses and in situ reference data ranged from -0.04 to -0.02 K and from 0.23 to 0.30 K respectively (the ranges are because different periods and reference data subsets yield small differences in results). Figure 1 and Table 1 show the results of the comparison of the dataset to in situ drifting buoy measurements. For the 1982-2024 period and for this reference dataset, the mean of the differences was -0.04 K and the standard deviation was 0.28 K. The dataset was found to have a stability over time of -0.0094 K per decade.

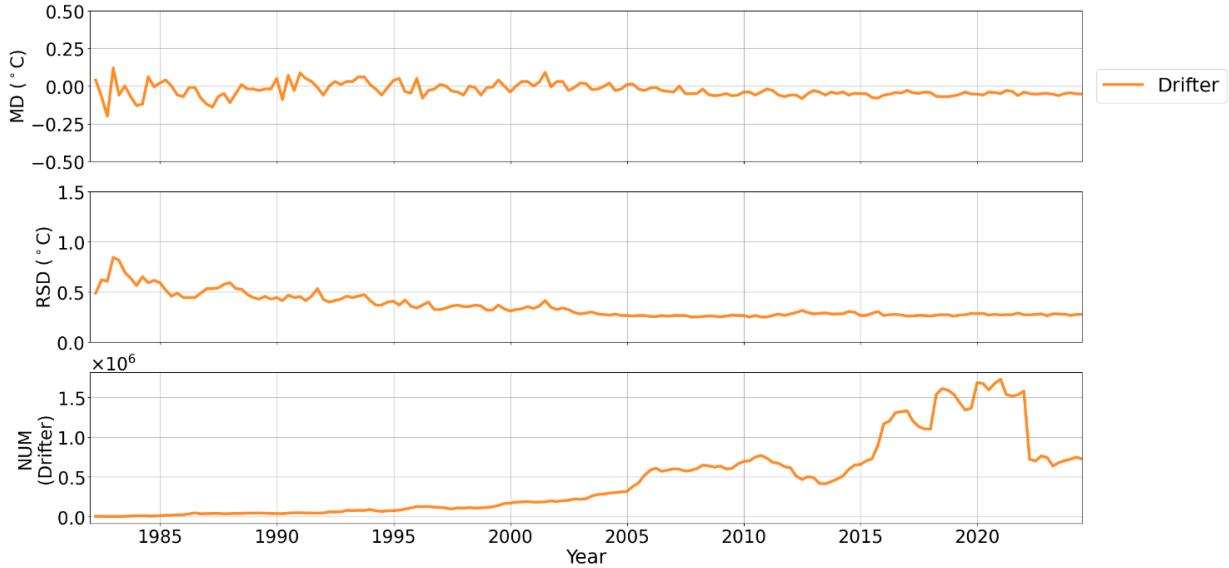

Figure 1. Timeseries of the C3S global L4 SST/IST validation results against in-situ observations from Drifters showing seasonal median differences (MD), Robust Standard Deviations (RSD) of the differences (middle panel) and the number of matchups (NUM, lower panel). The seasonal statistics are only calculated if >45 matchups are available.

Table 1. Validation statistical numbers for the full DMI C3S global L4 SST/IST time period against drifters.

Reference data	*Period*	*Difference*	*Robust Standard Deviation (RSD)*	*Number of match-ups*	*Stability, deg./decade*
Drifters	*1982 - 2024*	*-0.04*	*0.28*	*77580459*	*-0.010*

The time series of average SSTs were evaluated by comparing to the HadSST.4.1.1.0 dataset (Kennedy et al., 2019). The monthly anomaly data used to calculate the OMI were averaged to the same 5° grid resolution as HadSST. HadSST is generated from in situ data so there are gaps where there were no observations. Therefore, data were masked out so that both the HadSST and OMI datasets shared the same observational coverage. Then a global average time series was generated from each and the difference calculated.

Results are shown in Figure 2. There is an offset between the HadSST and OMI dataset time series because HadSST provides anomalies relative to the 1961-1990 average, when the ocean was cooler compared to the 1991-2020 average used for the OMI. The mean offset is -0.415 K. However, the first ~10 years of the time series has a smaller offset than the remainder. Since the early 1990s the HadSST time series and the time series generated from the dataset used for the OMI are relatively stable compared to each other. The standard deviation of the differences is 0.068K. It should be noted that these results reflect the uncertainty in both the HadSST data and the OMI data, and further investigation is required to understand the origin of the differences found here.

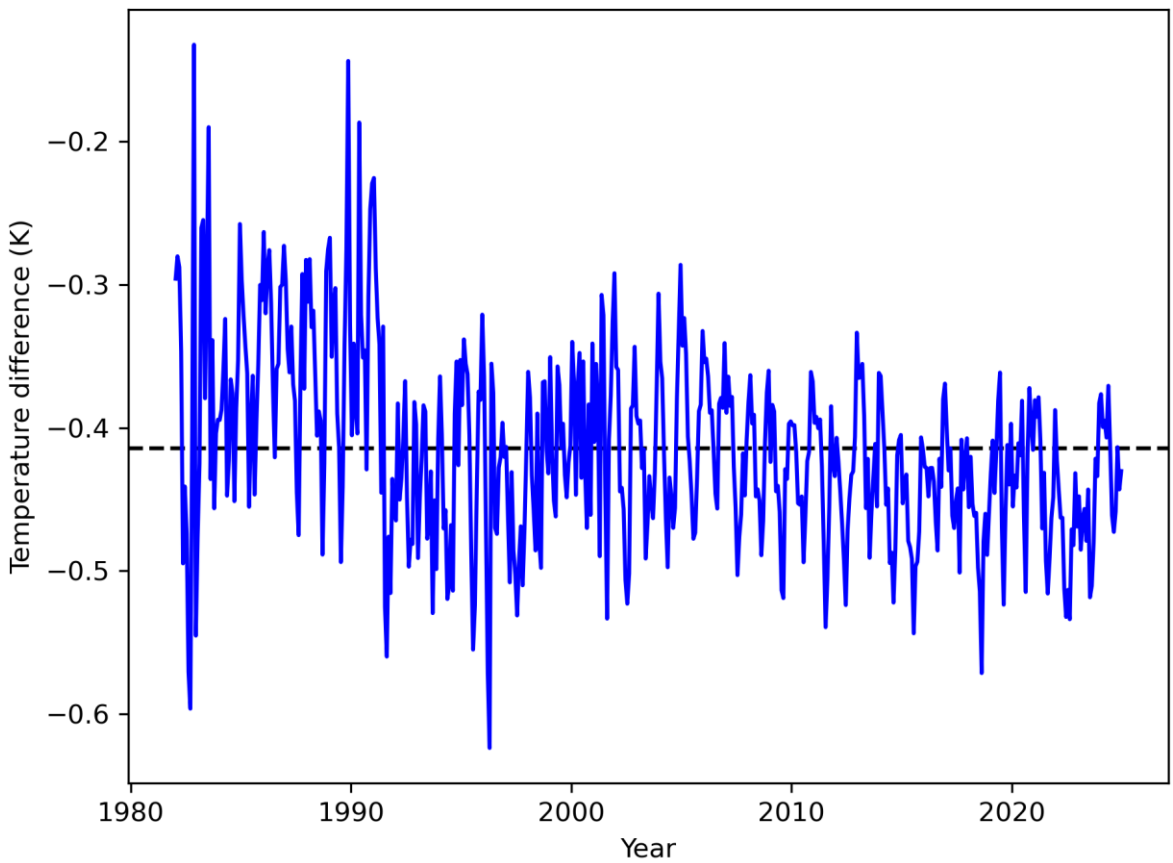

Figure 2. Difference between global average time series from the data used to generate the OMI and the HadSST dataset. The dashed horizontal line marks the mean difference between the two time series.

3) REFERENCES

Kennedy, J. J., Rayner, N. A., Atkinson, C. P., & Killick, R. E. (2019). An Ensemble Data Set of Sea Surface Temperature Change from 1850: The Met Office Hadley Centre HadSST.4.0.0.0 Data Set. Journal of Geophysical Research: Atmospheres, 124. https://doi.org/10.1029/2018JD029867

Pezzulli, S., Stephenson, D.B. and Hannachi A. 2005. The variability of seasonality. J Clim. 18: 71– 88, doi: 10.1175/JCLI-3256.1.

Sen, P.K. 1968. Estimates of the regression coefficient based on Kendall's tau. J Am Statist Assoc. 63: 1379– 1389, doi: 10.1080/01621459.1968.10480934.